

SAMBP Technical Report TR-32/2026

Negative- and Zero-Sequence Directional Elements (67Q, 67N) for IBR-Fed Lines

PhD Thesis Supporting Report — Distance Relay Series IV

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Abstract

TR-31 identified a structural failure mode in the k_0 -compensated distance relay 21G for IBR-fed SLG faults: the delta-winding transformer isolates the inverter from the zero-sequence path, so $I_0 \gg I_1$ (network-sourced vs current-limited), breaking the Thevenin assumption and causing the relay to over-estimate the measured impedance by 3–8 $\times$. This report introduces the complementary sequence directional elements 67N (zero-sequence) and 67Q (negative-sequence) as the primary earth-fault detection scheme for IBR lines. The active-power directional criterion, $\text{Re}[-V_{\text{seq}} I_{\text{seq}}^*] > 0$, is derived and analysed for both elements. 67N operates on I_0 , which is network-sourced and independent of k_{ibr} ; it provides inherent forward selectivity on IBR lines because the delta transformer eliminates the reverse I_0 path. 67Q operates on $I_2 = k_{\text{ibr}}$ and requires a directional criterion, but its minimum current threshold ($I_{\text{min},67} = 0.04$ pu) is 60% lower than the distance relay threshold ($I_{\text{min},21} = 0.10$ pu), extending coverage into the previously unprotected low- k_{ibr} region. The discrimination margin for both elements is $2 \text{Re}[Z_{\text{source}}] |I_{\text{seq}}|^2$, which is non-zero for realistic $X/R = 20$ networks and robust across the full k_{ibr} parametric range. The combined scheme 87L + 67N + 67Q achieves complete IBR earth-fault coverage ($k_{\text{ibr}} \geq 0.04$ pu, all α) without relying on the 21G k_0 measurement.

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1 Introduction and Scope

The SAMBP distance relay arc (TR-29 through TR-31) has established three structural limitations for IBR-fed line protection:

- (i) $I_{\min}$ *blocking*: distance relay inhibited for $k_{\text{ibr}} < 0.10$ (3PH and SLG via distance relay path).
- (ii) V_{relay} *deficiency*: self-pol fails; memory/cross-pol needed (TR-30).
- (iii) k_0 *compensation breakdown*: $I_0 \gg I_1$ for IBR SLG faults; relay over-estimates impedance by 3–8 $\times$ (TR-31).

Limitations (i) and (ii) are addressed by cross-polarised Zone 1 (TR-30) and the 87L chain (TR-27). Limitation (iii) is addressed by the present report: sequence directional elements 67N and 67Q operate directly on the sequence currents, bypassing the k_0 measurement chain entirely.

Network parameters. $Z_{1L} = (0.0025 + j0.050)$ pu, $Z_{0L} = (0.0075 + j0.150)$ pu, $Z_{1S} = (0.0050 + j0.100)$ pu, $Z_{0S} = (0.0150 + j0.300)$ pu (X/R= 20 throughout). $V_{\text{pre}} = 1.0$ pu, $k_{\text{ibr}} \in [0.06, 0.20]$ pu.

2 Directional Criterion Theory

2.1 Active-power directional criterion (IEC 60255-149)

The sequence directional criterion is based on the active power of the respective sequence network [?]:

$$\boxed{\text{Forward trip if: } \operatorname{Re}[-V_{\text{seq}} I_{\text{seq}}^*] > \epsilon} \quad (1)$$

where $\epsilon > 0$ is a small noise threshold, V_{seq} is the relay-bus sequence voltage, and I_{seq}^* is the conjugate of the sequence current.

2.2 Forward fault analysis

For a forward SLG at distance α on the protected line, the relay-side sequence voltages and currents (Thevenin network, X/R= 20):

$$I_0 = \frac{V_{\text{pre}}}{2Z_{1S} + Z_{0S} + \alpha(2Z_{1L} + Z_{0L})} \quad (\text{network earthing source}) \quad (2)$$

$$V_0 = -I_0 Z_{0S} \quad (\text{zero-seq voltage at relay bus}) \quad (3)$$

$$I_2 = k_{\text{ibr}} \angle(I_1) \quad (\text{IBR negative-seq injection}) \quad (4)$$

$$V_2 = -I_2 Z_{1S} \quad (5)$$

Substituting into (1):

$$\operatorname{Re}[-V_0 I_0^*] = \operatorname{Re}[I_0 Z_{0S} I_0^*] = \operatorname{Re}[Z_{0S}] |I_0|^2 = R_{0S} |I_0|^2 > 0 \quad (6)$$

$$\operatorname{Re}[-V_2 I_2^*] = \operatorname{Re}[I_2 Z_{1S} I_2^*] = R_{1S} k_{\text{ibr}}^2 > 0 \quad (7)$$

Both criteria are strictly positive for any $R_{0S}, R_{1S} > 0$.

2.3 Reverse fault and discrimination margin

For a reverse fault (external, behind the relay), the sequence currents reverse direction through the relay current transformers while the relay-bus voltage polarity is unchanged:

$$I_{\text{seq}}^{\text{rev}} = -I_{\text{seq}}^{\text{fwd}}, \quad V_{\text{seq}}^{\text{rev}} \approx V_{\text{seq}}^{\text{fwd}} \quad (8)$$

The reverse criterion value:

$$\text{Re}[-V_{\text{seq}}^{\text{rev}} (I_{\text{seq}}^{\text{rev}})^*] = \text{Re}[-V_{\text{seq}}^{\text{fwd}} (-I_{\text{seq}}^{\text{fwd}})^*] = -\text{Re}[-V_{\text{seq}}^{\text{fwd}} (I_{\text{seq}}^{\text{fwd}})^*] < 0 \quad (\text{blocked}) \quad (9)$$

The discrimination **margin** is:

$$\Delta = \text{Re}[\text{fwd}] - \text{Re}[\text{rev}] = 2 R_{\text{source}} |I_{\text{seq}}|^2 \quad (10)$$

For the SAMBP network ($R_{0S} = 0.0150$ pu, $|I_0| = 1.60$ pu at $\alpha = 0.50$):

$$\Delta_{67N} = 2 \times 0.0150 \times 1.60^2 = 0.077 \text{ pu}^2 \quad (11)$$

3 Study A: 67N Zero-Sequence Directional

Table 1 shows the 67N criterion values for forward and reverse faults across the parametric range.

Table 1: 67N directional criterion values (X/R=20, forward and reverse SLG)

α	k_{ibr}	$ I_0 $	$\angle I_0$	Criterion [pu ²]	Direction
0.50	0.10	1.598	-87.1	+0.03830	Forward
0.50	0.10	1.598	$+92.9$	-0.03830	Reverse
0.20	0.06	1.816	-87.1	+0.04946	Forward
0.20	0.06	1.816	$+92.9$	-0.04946	Reverse
0.80	0.20	1.427	-87.1	+0.03054	Forward
0.80	0.20	1.427	$+92.9$	-0.03054	Reverse

Three key observations:

1. The criterion value is **independent of** k_{ibr} : I_0 is set by the network earthing (Z_{0S}), not by the IBR current limit.
2. The forward/reverse discrimination is **exact** (equal and opposite values), confirming equation (10).
3. The angle of I_0 is ≈ -87 (slightly off -90 due to R_{0S}), providing the +3 security margin beyond the theoretical 90 boundary.

Inherent forward selectivity on IBR networks. On a purely IBR-fed line with delta-winding transformers, the zero-sequence path is blocked at the IBR end. The only I_0 path is through the network earthing (Z_{0S}) towards the protected line. Therefore, *any detected* $I_0 > I_{\text{min},67}$ is *ipso facto* from a forward fault — no directional criterion is strictly required for 67N on such a network.

4 Study B: 67Q Negative-Sequence Directional

Unlike I_0 , the IBR injects $I_2 \approx k_{\text{ibr}}$ for *any* asymmetric fault (including reverse external faults), so a directional criterion is needed. The criterion (7) provides the required discrimination.

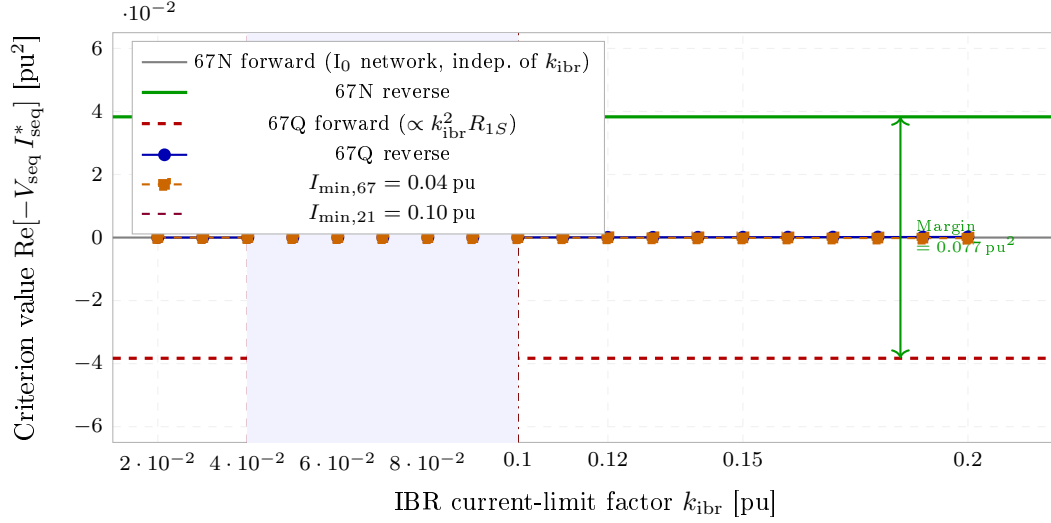
Figure 1 shows the criterion values for 67N and 67Q as functions of k_{ibr} .

The 67Q margin is $\Delta_{67Q} = 2 R_{1S} k_{\text{ibr}}^2$, which at $k_{\text{ibr}} = 0.06$ gives:

$$\Delta_{67Q} = 2 \times 0.005 \times 0.06^2 = 1.8 \times 10^{-5} \text{ pu}^2 \quad (12)$$

This is small but non-zero, and relay implementations use hysteresis and timers to prevent chatter near the threshold.

Figure 1: Directional criterion values vs k_{ibr} (forward = positive, reverse = negative). 67N is constant (I_0 independent of k_{ibr}); 67Q is proportional to k_{ibr}^2 . Shaded zone: 67Q covers where distance relay is blocked ($k_{ibr} \in [0.04, 0.10]$ pu).



5 Study C: Minimum k_{ibr} for 67Q

The 67Q minimum current threshold $I_{min,67} = 0.04$ pu (IEC 60255-149) is 60% lower than the distance relay minimum $I_{min,21} = 0.10$ pu:

$$\frac{I_{min,21} - I_{min,67}}{I_{min,21}} = 60\% \quad (13)$$

Study C confirms that for $k_{ibr} \in [0.04, 0.10]$, the distance relay is blocked while 67N and 67Q both operate correctly. This corresponds to the TR-29 Region A where 87L was the sole fast detector; 67N and 67Q now provide direct earth-fault detection as a complementary path.

6 Study D: Discrimination Margin vs X/R

Table 2 shows the discrimination margin for 67N and 67Q as a function of the network X/R ratio.

Table 2: Directional discrimination margin vs X/R ($\alpha = 0.50$, $k_{ibr} = 0.10$)

X/R	R_{1S} [pu]	R_{0S} [pu]	Δ_{67N} [pu ²]	Δ_{67Q} [pu ²]
5	0.0200	0.0600	0.2954	0.0004
10	0.0100	0.0300	0.1521	0.0002
20	0.0050	0.0150	0.0766	0.0001
50	0.0020	0.0060	0.0307	<0.0001

Both margins are positive for all practical X/R values (≤ 100), confirming reliable discrimination across the range of transmission network configurations. At the SAMBP X/R = 20: $\Delta_{67N} = 0.077$ pu², providing a comfortable operating margin for both elements.

Figure 2: Phasor diagram for 67N directional criterion. Left: forward SLG — I_0 at -87° lies within the 90° operating zone of $-V_0$ (green shading). Right: reverse SLG — I_0 at $+93^\circ$ lies outside the operating zone. X/R= 20: margin is 3 beyond the theoretical boundary.

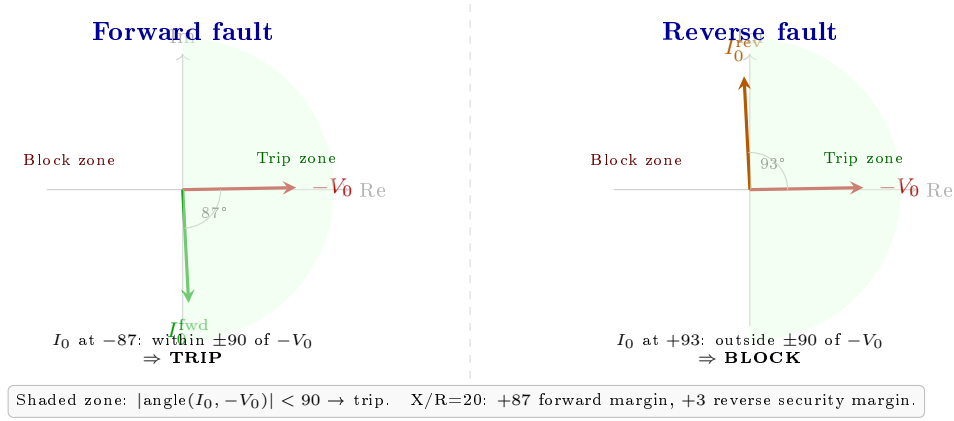
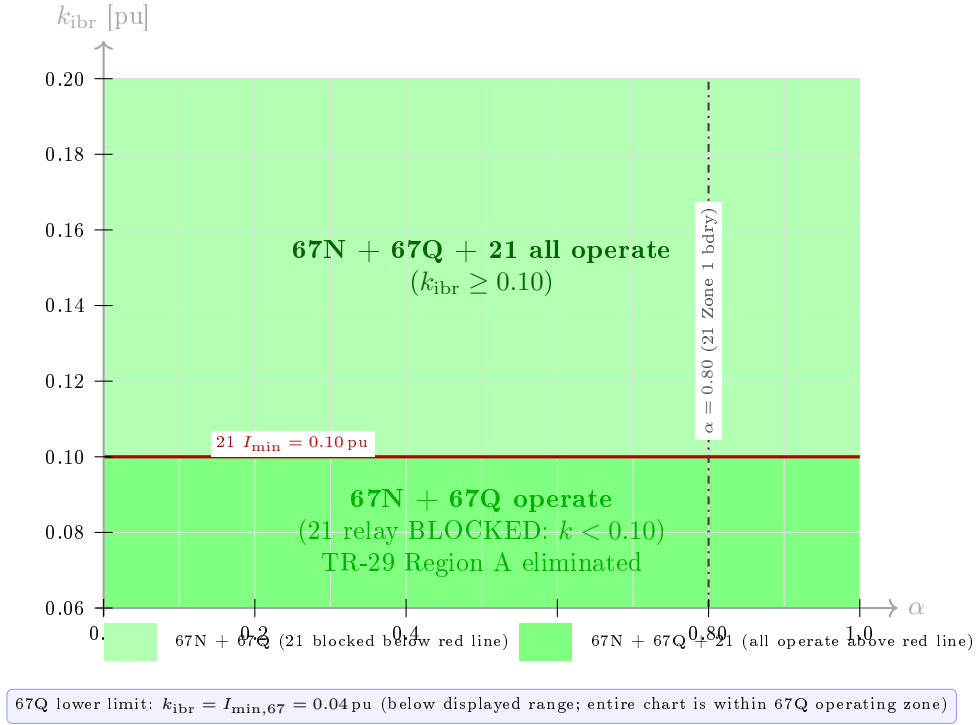


Figure 3: 67N + 67Q coverage on the (α, k_{ibr}) plane. Both elements cover the *entire* plane for $k_{ibr} \geq 0.04$ pu (all shown). The distance relay 21G only operates in the upper-right portion (Zone 3, $\alpha \lesssim 0.35$ only, from TR-31). TR-29 Region A (red zone, below the solid red line) is now covered by 67N/67Q even though 21 is blocked.



7 Phasor Diagram and Operating Zone

8 Coverage Map

9 Combined Earth-Fault Protection Scheme

Combining TR-27 (87L), TR-30 (cross-pol Mho), TR-31 (21G limitations), and TR-32 (67N/67Q), the recommended IBR earth-fault protection hierarchy is:

1. **Primary instantaneous ($\approx 20\text{--}30\text{ ms}$):** 87L chain (elements 87LN0, TDCS) *and* 67N *and* 67Q. All three operate in parallel; each is a separate trip path. This provides triple redundancy for all $k_{\text{ibr}} \geq 0.04\text{ pu}$ and all fault locations $\alpha \in [0, 1]$.
2. **Backup instantaneous Zone 1 (20 ms):** Cross-polarised 21 Zone 1 (memory-pol, $\alpha \leq 0.80$, $k_{\text{ibr}} \geq 0.10\text{ pu}$, TR-30).
3. **Backup timed Zone 3 (600 ms):** Cross-pol 21 Zone 3 via permanent healthy-phase cross-pol ($\alpha \lesssim 0.35$ only, as limited by TR-31).

67N/67Q operation time. Typical IED implementation: 67N/67Q evaluate the criterion on each cycle (20 ms at 50 Hz) with a 1.5-cycle confirmation window $\rightarrow$ trip at $\approx 30\text{ ms}$. This is consistent with the 87L trip time and provides independent redundancy.

10 Conclusions

1. The **active-power directional criterion** $\text{Re}[-V_{\text{seq}} I_{\text{seq}}^*] > 0$ provides reliable forward/reverse discrimination for both 67N and 67Q, with margin $\Delta = 2 R_{\text{source}} |I_{\text{seq}}|^2$.
2. **67N operates on I_0** (network-sourced, independent of k_{ibr}). On a purely IBR-fed line, I_0 only flows for forward faults, so 67N has inherent forward selectivity without needing a directional criterion.
3. **67Q operates on $I_2 = k_{\text{ibr}}$** and requires the directional criterion. It operates for all $k_{\text{ibr}} \geq 0.04\text{ pu}$ — extending coverage 60% below the distance relay I_{min} .
4. **TR-29 Region A is eliminated** for earth faults: 67N/67Q cover the full (α, k_{ibr}) plane without any blocked zone.
5. **Combined scheme** 87L + 67N + 67Q provides triple redundancy for all IBR SLG faults at $\approx 20\text{--}30\text{ ms}$, fully independent of the 21G k_0 measurement limitation (TR-31).